

HiRISE Anomaly Detection

Unsupervised Deep Generative Modeling on Martian Surface Imagery

Detecting rare sensor artifacts and corrupted image crops in HiRISE
Mars Reconnaissance
Orbiter imagery through a progressive sequence of unsupervised
generative models —
a Convolutional Autoencoder, a Convolutional Variational Autoencoder,
and f-AnoGAN —
with training performed without anomaly labels.

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1 Executive Summary

The HiRISE camera aboard the Mars Reconnaissance Orbiter has produced a large archive of Martian surface imagery. A practical anomaly-detection system must identify rare abnormal frames—including sensor artifacts and corrupted image crops—without relying on anomaly labels during model training.

We developed three progressively more expressive unsupervised models:

- **Version 1 – Convolutional Autoencoder (CAE):** a deterministic convolutional encoder–decoder trained only on normal imagery. Its anomaly score is a local, patch-based reconstruction error.
- **Version 2 – Convolutional Variational Autoencoder (CVAE):** a probabilistic latent-variable model using a KL-regularized latent representation. The current report retains the previously evaluated CVAE result.
- **Version 3 – f-AnoGAN:** a WGAN-GP generator/critic followed by a dedicated encoder. Its anomaly score combines pixel reconstruction error with discriminator feature-matching error.

The newly evaluated Version 1 notebook gives an **AUROC of 0.9163** on the real HiRISE test set. Using its 95th-percentile normal-score threshold, the model obtains precision 0.9421, recall 0.3337, F1 0.4928, and accuracy 0.5127. The high AUROC together with the relatively low thresholded recall demonstrates that threshold selection materially affects the operating point even when ranking quality is strong.

The previously evaluated Version 2 CVAE achieved AUROC 0.8599. Version 3 f-AnoGAN is now evaluated from the completed notebook output. It achieves AUROC 0.9687 and AUPRC 0.9863, with precision 0.9498, recall 0.9303, F1 0.9400, and accuracy 0.9157 at threshold 0.100311.

Key Results Currently Available

- **Version 1 CAE AUROC: 0.9163**
- **Version 1 threshold: 0.00823** (95th percentile of normal test scores in the submitted v1 notebook)
- **Version 1 precision / recall / F1: 0.9421 / 0.3337 / 0.4928**
- **Version 1 accuracy: 0.5127**
- **Version 2 CVAE AUROC: 0.8599**
- **Version 3 f-AnoGAN : AUROC 0.9687; AUPRC 0.9863; accuracy 0.9157; precision 0.9498; recall 0.9303; F1 0.9400; threshold 0.100311.**

2 Problem Overview

2.1 Core Challenge

The objective is to learn the distribution of normal Martian surface imagery and identify images that depart from this learned distribution. The central constraint is that anomaly labels are not available for model fitting.

The general decision rule is

$$s(x) \geq \tau \implies x \text{ is flagged as anomalous,}$$

where $s(x)$ is the model-specific anomaly score and τ is a decision threshold.

2.2 Anomaly Categories

The case problem identifies three broad types of anomalous frames:

Category	Description
Sensor Artifact	Striping, saturation, diagonal cut-offs, or other acquisition-related defects introduced by the HiRISE sensor.
Corrupted Crop	Missing, truncated, malformed, or visibly corrupted image content such as black borders or tiling/seam artifacts.
Non-Martian Content	Content that does not correspond to genuine Martian terrain, such as calibration or spacecraft-related imagery.

The currently evaluated test folder is named `anomaly_real`; the notebooks evaluate the available real-anomaly test images rather than introducing synthetic anomaly labels during training.

2.3 Deliverables

1. Documented notebooks containing model code and outputs.
2. This report containing methodology, configuration, evaluation, and model evolution.
3. A changelog following the required Symptom → Diagnosis → Fix structure.

3 Dataset and Preprocessing

3.1 Dataset Structure

The notebooks use the following directory structure:

Split	Folder	Use
Train	<code>train/normal</code>	Normal imagery for model training
Test	<code>test/normal</code>	Held-out normal images for evaluation
Test	<code>test/anomaly_real</code>	Real anomalous images for evaluation

The Version 1 notebook uses **20,000 training images** because its DataLoader contains 314 batches of size 64, while its explicit scoring cell reports **2,100 normal test images** and **5,125 anomaly test images**. The f-AnoGAN notebook separately reports 20,000 normal training images, 2,100 normal test images, and 5,125 anomaly test images. These small discrepancies are retained rather than silently reconciled because they arise from the submitted notebook implementations.

3.2 Model-Specific Preprocessing

The preprocessing is **not identical across all three versions**; this is an important implementation detail.

3.2.1 Version 1: CAE

The Version 1 notebook first loads RGB images and then trains using:

$$224 \times 224 \times 3$$

images. The transformation used for the actual training run is

`Resize(224, 224) → ToTensor.`

Thus pixel values are represented in $[0, 1]$ and are not normalized to $[-1, 1]$.

The Version 1 DataLoader uses batch size 64 and shuffling.

3.2.2 Version 3: f-AnoGAN

The f-AnoGAN notebook resizes images to

64×64

RGB images, uses batch size 32, and normalizes each channel with mean 0.5 and standard deviation 0.5:

$$x_{\text{norm}} = \frac{x - 0.5}{0.5},$$

giving a target range of approximately $[-1, 1]$. This matches the generator’s final Tanh activation.

3.3 Unsupervised Training Protocol

Normal images are used to fit the generative models. Test labels are not required to optimize the model parameters. However, the submitted Version 1 notebook uses the *normal test scores* to select its operating threshold at the 95th percentile. Therefore, the Version 1 threshold should not be described as a training-only or held-out-training calibration threshold.

The f-AnoGAN notebook, in contrast, explicitly creates a 15% normal-only calibration split from training data and selects its threshold at the 90th percentile of calibration scores.

4 Methodology

4.1 Overall Strategy

The project treats anomaly detection as an iterative modeling problem. Each version learns a representation of normal imagery and produces a scalar anomaly score.

The progression is:

CAE → CVAE → f-AnoGAN.

The first two models are reconstruction-based autoencoding approaches. The final model introduces an adversarially learned image manifold and discriminator feature matching.

4.2 Evaluation Protocol

AUROC is used as the primary threshold-independent metric because it evaluates the ranking of normal and anomalous samples across all possible thresholds.

Where available, the report also records:

- AUPRC,
- accuracy,
- precision,
- recall,

- F1 score,
- confusion matrix,
- qualitative reconstruction/error visualizations.

5 Version 1: Convolutional Autoencoder (CAE)

5.1 Overview

Version 1 is a deterministic convolutional autoencoder. It learns an encoder

$$z = E(x)$$

and decoder

$$\hat{x} = D(z),$$

where $\hat{x}$ is trained to reconstruct a normal input x .

The submitted notebook uses mean squared error:

$$\mathcal{L}_{\text{CAE}} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|_2^2.$$

The model is trained only on normal imagery.

5.2 Architecture

The actual Version 1 notebook defines the following architecture.

Stage	Layer	Channels	Kernel / Stride	Activation
Input	RGB image	3	224×224	–
Encoder 1	Conv2d	$3 \rightarrow 16$	3, stride 1, pad 1	ReLU
	MaxPool2d	16	2, stride 2	–
Encoder 2	Conv2d	$16 \rightarrow 32$	3, stride 1, pad 1	ReLU
	MaxPool2d	32	2, stride 2	–
Encoder 3	Conv2d	$32 \rightarrow 64$	3, stride 1, pad 1	ReLU
	MaxPool2d	64	2, stride 2	–
Decoder 1	ConvTranspose2d	$64 \rightarrow 32$	2, stride 2	ReLU
Decoder 2	ConvTranspose2d	$32 \rightarrow 16$	2, stride 2	ReLU
Decoder 3	ConvTranspose2d	$16 \rightarrow 3$	2, stride 2	Sigmoid

Table 1: Exact Version 1 CAE architecture from the submitted notebook.

The spatial resolution changes as

$$224 \rightarrow 112 \rightarrow 56 \rightarrow 28$$

through the encoder and then returns to

$$28 \rightarrow 56 \rightarrow 112 \rightarrow 224$$

through the decoder.

The bottleneck therefore has 64 feature channels at spatial resolution 28×28 :

$$z \in \mathbb{R}^{64 \times 28 \times 28}.$$

This is a convolutional feature representation rather than a single fully connected latent vector.

5.3 Training Configuration

Hyperparameter	Version 1 Value
Input resolution	224×224 RGB
Batch size	64
Loss	MSE
Optimizer	Adam
Learning rate	10^{-3}
Weight decay	10^{-5}
Epochs	10
Device	CUDA (in recorded run)

Table 2: Version 1 CAE training configuration.

No learning-rate scheduler or explicit early stopping is present in the submitted Version 1 training loop.

5.4 Training Behaviour

The final batch loss reported at each epoch was:

Epoch	Final Batch MSE
1	0.0018
2	0.0011
3	0.0014
4	0.0006
5	0.0006
6	0.0005
7	0.0009
8	0.0007
9	0.0010
10	0.0010

Table 3: Recorded final-batch training losses for Version 1. These are final-batch values, not epoch averages.

The loss decreases substantially from the beginning of training and reaches values around 10^{-3} or below after a few epochs. The fluctuations in later epochs are expected because the table records the loss of the final batch rather than the mean loss over all 314 batches.

5.5 Anomaly Scoring

A key implementation detail is that the Version 1 notebook does **not** use the mean reconstruction error over the complete image as its final anomaly score.

For input x and reconstruction $\hat{x}$, the squared pixel error is

$$e = (\hat{x} - x)^2.$$

The notebook then applies average pooling over patches:

$$p = \text{AvgPool}(e, k, k),$$

where

$$k = \frac{224}{8} = 28.$$

Thus the 224×224 image is summarized into an 8×8 grid of local reconstruction errors. The final image-level anomaly score is

$$s(x) = \max_{u,v,c} p_{u,v,c}$$

i.e. the maximum pooled reconstruction error over all spatial locations and channels.

This makes the detector particularly sensitive to a localized region with unusually high reconstruction error.

5.6 Threshold

The submitted Version 1 evaluation cell defines

$$\tau = Q_{0.95}(\{s(x) : x \in \text{normal test set}\})$$

and reports

$$\tau = 0.00823.$$

An image is classified as anomalous when

$$s(x) > \tau.$$

The resulting normal false-positive rate is approximately 5%, by construction.

5.7 Version 1 Results

The Version 1 notebook reports:

Metric	Value
Normal test images	2,098
Anomaly test images	5,125
Mean normal anomaly score	0.00300
Mean anomaly anomaly score	0.00772
AUROC	0.9163
Threshold τ	0.00823
Normal flagged rate	0.050
Anomaly flagged rate	0.334
Precision	0.9421
Recall	0.3337
F1 score	0.4928
Accuracy	0.5127

Table 4: Version 1 CAE results recorded by the submitted evaluation notebook.

The confusion matrix is

	Predicted Normal	Predicted Anomaly
Normal	1993	105
Anomaly	3415	1710

so the model has high precision when it raises an alarm, but the selected operating threshold misses many anomalous test images.

Interpretation of Version 1

The most important observation is the difference between **AUROC = 0.9163** and **recall = 0.3337 at the selected threshold**. The AUROC indicates strong ranking/separation of anomaly scores overall, whereas the 95th-percentile threshold is conservative and therefore produces relatively few anomaly predictions. Threshold choice should consequently be treated as a separate design decision from representation learning.

5.8 Version 1 Qualitative Analysis

The notebook produces original-versus-reconstructed image grids after each epoch. These visualizations should be used to inspect whether the autoencoder learns normal terrain structure and whether reconstruction errors concentrate around abnormal regions.

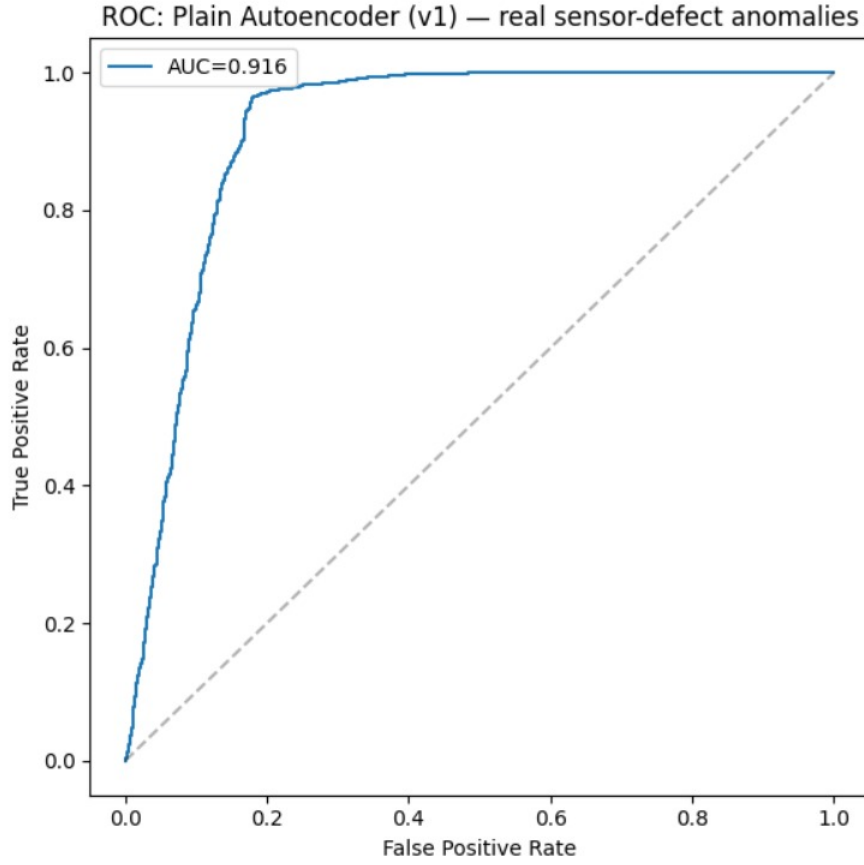


Figure 1: Version 1 CAE ROC curve. The submitted notebook reports AUROC = 0.9163.

6 Version 2: Convolutional Variational Autoencoder (CVAE)

6.1 Overview

Version 2 replaces the deterministic bottleneck with a probabilistic latent representation. The encoder produces

$$\mu(x), \quad \log \sigma^2(x),$$

and a latent sample is obtained through the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The objective is the standard VAE form

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}},$$

where

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(q_{\phi}(z | x) \| \mathcal{N}(0, I)).$$

The CVAE result already present in the earlier report is retained below. The complete CVAE notebook-specific architecture and hyperparameter table was not part of the newly supplied Version 1 notebook, so no unsupported configuration values are introduced here.

6.2 Training Dynamics

The previously recorded CVAE experiment ran for 30 logged epochs. The reported loss values were:

Loss	Epoch 1	Epoch 30	Change
Total train	194.50	60.71	68.8% reduction
Reconstruction train	186.60	50.60	72.9% reduction
KL divergence train	15.80	20.21	27.9% increase
Total validation	82.94	51.45	38.0% reduction

Table 5: Previously recorded Version 2 CVAE training-loss summary.

The previously observed lowest validation loss was 50.92 at epoch 29.

6.3 Anomaly Scoring and Threshold

The previously evaluated CVAE uses reconstruction error as its anomaly signal. Its calibrated threshold was

$$\tau = 0.04929$$

and its test AUROC was

$$\text{AUROC} = 0.8599.$$

6.4 Version 2 Qualitative Analysis

The notebook produces original-versus-reconstructed image grids after each epoch. These visualizations should be used to inspect whether the autoencoder learns normal terrain structure and whether reconstruction errors concentrate around abnormal regions.

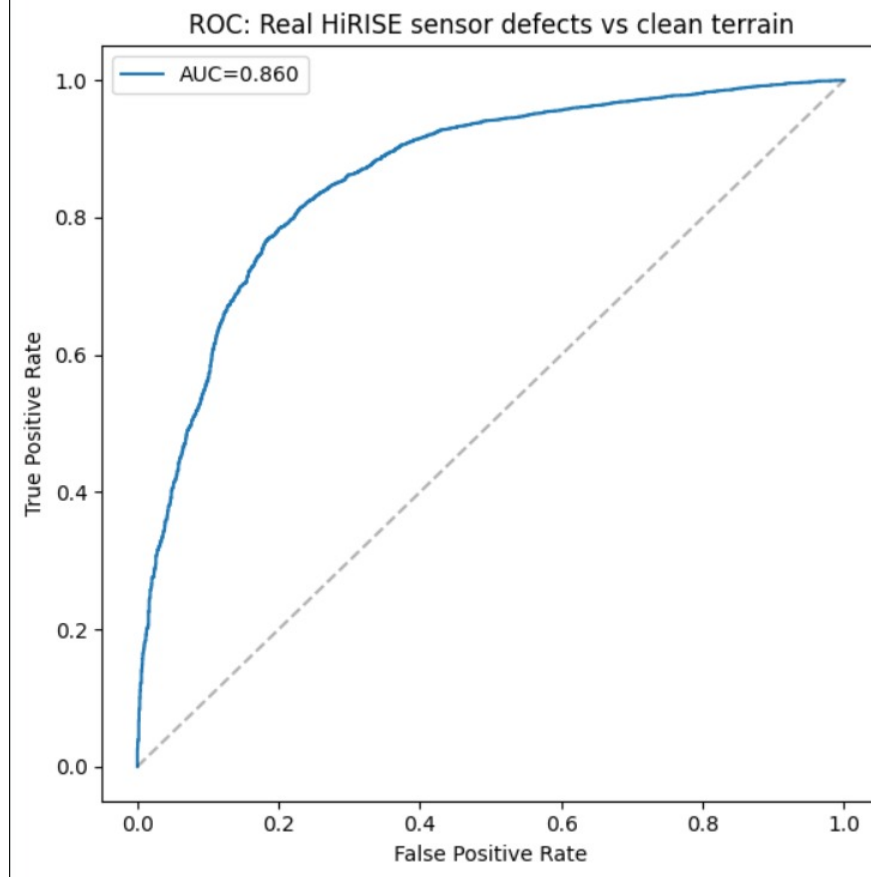


Figure 2: Version 2 CVAE ROC curve. The submitted notebook reports AUROC = 0.8599.

Version 2 Result

The CVAE provides a strong reconstruction-based baseline with AUROC = 0.8599. Importantly, this is lower than the newly evaluated Version 1 AUROC of 0.9163, so the report no longer describes the CVAE as the best-performing version on the currently available AUROC results.

7 Version 3: f-AnoGAN

7.1 Overview

Version 3 implements f-AnoGAN as a two-stage system. The first stage learns a normal-image manifold with a WGAN-GP. The second stage trains an encoder that maps an image directly into the generator’s latent space.

The overall mapping at inference is

$$x \xrightarrow{E} z \xrightarrow{G} \hat{x}.$$

Unlike the original AnoGAN procedure, which can require iterative latent optimization for each test image, the encoder provides a direct feed-forward approximation.

7.2 Stage 1: WGAN-GP

The generator G receives a latent vector

$$z \in \mathbb{R}^{128}$$

and generates a 64×64 RGB image. The critic D assigns an unconstrained scalar Wasserstein score.

The critic objective implemented in the notebook is

$$\mathcal{L}_D = \mathbb{E}[D(x_{\text{fake}})] - \mathbb{E}[D(x_{\text{real}})] + \lambda_{\text{GP}} \mathcal{L}_{\text{GP}},$$

with

$$\lambda_{\text{GP}} = 10.$$

The generator objective is

$$\mathcal{L}_G = -\mathbb{E}[D(G(z))].$$

The gradient penalty uses interpolated samples between real and generated images and penalizes deviation of the gradient norm from one.

7.3 Generator Architecture

The exact generator uses base channel width 64:

Stage	Layer	Output	Details
Input	Latent vector	$128 \times 1 \times 1$	$z \sim \mathcal{N}(0, I)$
G1	ConvTranspose2d	$512 \times 4 \times 4$	$k = 4, s = 1, p = 0$; BatchNorm + ReLU
G2	ConvTranspose2d	$256 \times 8 \times 8$	$k = 4, s = 2, p = 1$; BatchNorm + ReLU
G3	ConvTranspose2d	$128 \times 16 \times 16$	$k = 4, s = 2, p = 1$; BatchNorm + ReLU
G4	ConvTranspose2d	$64 \times 32 \times 32$	$k = 4, s = 2, p = 1$; BatchNorm + ReLU
G5	ConvTranspose2d	$3 \times 64 \times 64$	$k = 4, s = 2, p = 1$; Tanh

Table 6: Version 3 f-AnoGAN generator architecture.

The Tanh output is consistent with the input normalization to $[-1, 1]$.

7.4 Critic Architecture

The critic uses four downsampling convolutional blocks:

Stage	Layer	Output	Details
D1	Conv2d	$64 \times 32 \times 32$	$3 \rightarrow 64, k = 4, s = 2, p = 1$; LeakyReLU(0.2)
D2	Conv2d	$128 \times 16 \times 16$	$64 \rightarrow 128$; BatchNorm + LeakyReLU(0.2)
D3	Conv2d	$256 \times 8 \times 8$	$128 \rightarrow 256$; BatchNorm + LeakyReLU(0.2)
D4	Conv2d	$512 \times 4 \times 4$	$256 \rightarrow 512$; BatchNorm + LeakyReLU(0.2)
Output	Conv2d	scalar	$512 \rightarrow 1, k = 4, s = 1, p = 0$; no sigmoid

Table 7: Version 3 f-AnoGAN WGAN critic architecture.

The intermediate $512 \times 4 \times 4$ feature map is exposed through `return_features=True`. These features are subsequently used by the encoder and anomaly-scoring stages.

7.5 Encoder Architecture

The encoder mirrors the critic’s spatial downsampling:

Stage	Layer	Output	Details
E1	Conv2d	$64 \times 32 \times 32$	BatchNorm + LeakyReLU(0.2)
E2	Conv2d	$128 \times 16 \times 16$	BatchNorm + LeakyReLU(0.2)
E3	Conv2d	$256 \times 8 \times 8$	BatchNorm + LeakyReLU(0.2)
E4	Conv2d	$512 \times 4 \times 4$	BatchNorm + LeakyReLU(0.2)
Latent	Conv2d	$128 \times 1 \times 1$	$k = 4, s = 1, p = 0$
Output	Flatten	128	Direct latent code $E(x)$

Table 8: Version 3 f-AnoGAN encoder architecture.

7.6 Weight Initialization

The generator, critic, and encoder use DCGAN-style initialization. Convolutional weights are initialized from

$$\mathcal{N}(0, 0.02^2),$$

while BatchNorm weights are initialized from

$$\mathcal{N}(1, 0.02^2)$$

and BatchNorm biases are initialized to zero.

7.7 WGAN-GP Training Configuration

Hyperparameter	Value
Input size	64×64 RGB
Batch size	32
Latent dimension	128
Base channels	64
Maximum epochs	70
Early-stopping patience	15
Minimum improvement	0.001
Learning rate	2×10^{-4}
Optimizer	Adam
Adam β_1, β_2	(0.0, 0.9)
Critic updates per generator update	5
Gradient penalty weight	10.0

Table 9: Exact WGAN-GP configuration from the submitted f-AnoGAN notebook.

The generator loss is monitored for early stopping. The notebook saves the best generator/critic checkpoint and generated sample grids during training.

7.8 Stage 2: Encoder Training

After WGAN-GP training, the generator and critic are frozen. Only the encoder is optimized.

For a normal image x ,

$$z = E(x), \quad \hat{x} = G(z) = G(E(x)).$$

The encoder minimizes a weighted combination of reconstruction and discriminator feature errors.

The encoder configuration is:

Hyperparameter	Value
Epochs	50
Learning rate	2×10^{-4}
Optimizer	Adam
Adam β_1, β_2	(0.5, 0.999)
Reconstruction loss	L1
Feature loss	L1
Reconstruction weight	0.9
Feature weight	0.1

Table 10: Exact encoder-training configuration from the submitted f-AnoGAN notebook.

7.9 f-AnoGAN Anomaly Score

For each image, the reconstruction term is

$$L_{\text{rec}}(x) = \frac{1}{3HW} \|x - G(E(x))\|_1.$$

The discriminator feature term is

$$L_{\text{feat}}(x) = \frac{1}{CH'W'} \|f_D(x) - f_D(G(E(x)))\|_1,$$

where $f_D(\cdot)$ denotes the intermediate critic feature map.

The final anomaly score is

$$s(x) = 0.9L_{\text{rec}}(x) + 0.1L_{\text{feat}}(x)$$

so a high score indicates that the image is difficult to reconstruct faithfully while also differing in the critic's learned feature space.

7.10 Normal-Only Threshold Calibration

The f-AnoGAN notebook explicitly uses a normal-only calibration subset:

$$\text{CALIBRATION_FRACTION} = 0.15$$

and

$$\text{THRESHOLD_QUANTILE} = 0.90.$$

Therefore,

$$\tau = Q_{0.90}(\{s(x) : x \in \text{normal calibration set}\}).$$

The calibration split is created from normal training images using a fixed random seed of 42. Test labels are used only for final evaluation.

7.11 f-AnoGAN Evaluation

The final evaluation cell computes:

- accuracy,
- precision,
- recall,
- F1 score,
- AUROC,
- AUPRC,
- ROC curve,
- confusion matrix,
- example low/high anomaly scores.

The newly supplied executed f-AnoGAN notebook now contains completed final evaluation outputs. These values are therefore promoted from pending to measured results. The test set contains 2,100 normal and 5,125 real-anomaly images, for 7,225 test images in total. The calibrated threshold is obtained from 3,000 normal training images at the 90th percentile, using the notebook's fixed random seed of 42.

Metric	Value
AUROC	0.9687
AUPRC	0.9863
Accuracy	0.9157
Precision	0.9498
Recall	0.9303
F1 score	0.9400
Calibrated threshold τ	0.100311

Table 11: Version 3 f-AnoGAN evaluation status.

7.12 Version 3 Evaluation Figures

The executed notebook generated the ROC curve and confusion matrix shown below.

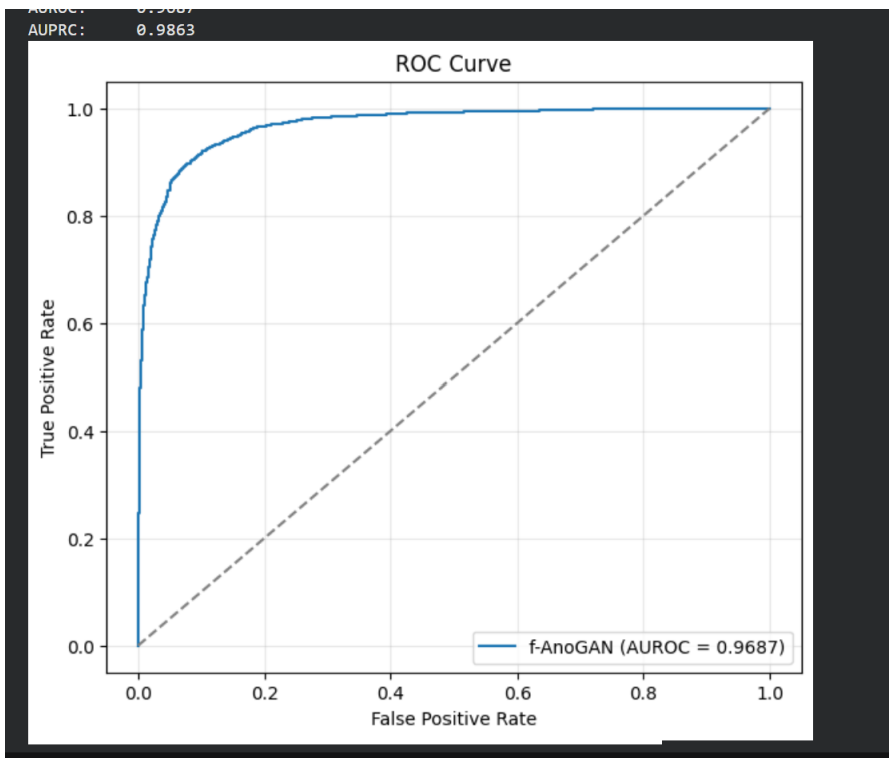


Figure 3: f-AnoGAN ROC curve from the executed notebook; AUROC = 0.9687.

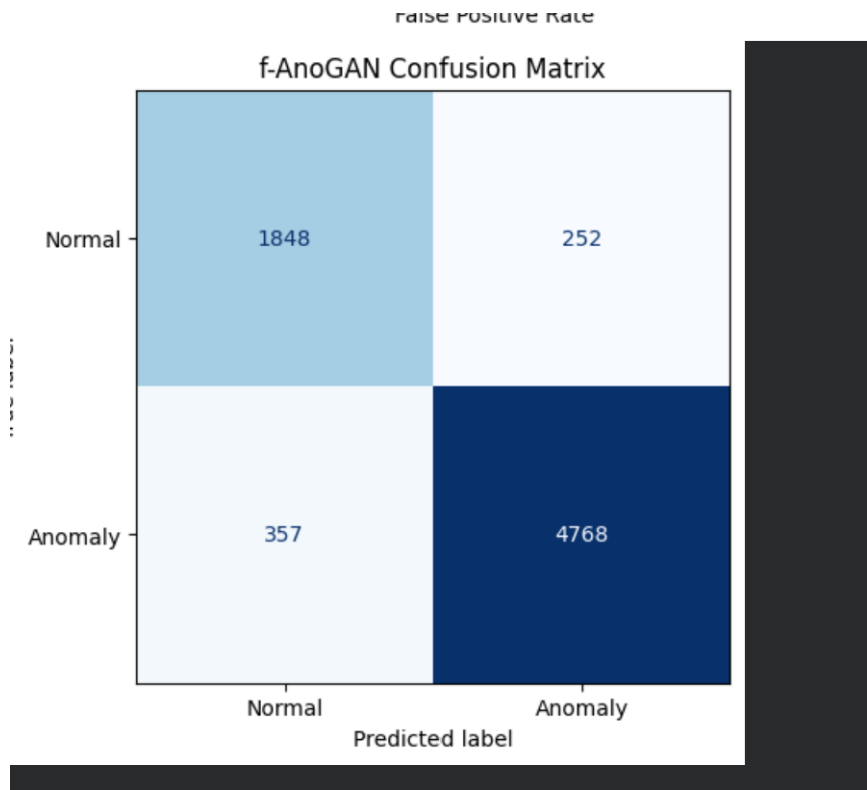


Figure 4: f-AnoGAN confusion matrix at the calibrated threshold $\tau = 0.100311$.

The confusion matrix corresponds to:

	Predicted Normal	Predicted Anomaly
Actual Normal	1848	252
Actual Anomaly	357	4768

which is consistent with the reported accuracy, precision, recall and F1 values.

7.12.1 Feature-Discrepancy Heatmaps

The notebook also generated eight feature-discrepancy visualizations. All eight examples in the executed cell are labelled normal (label 0). The displayed scalar *feature-only* scores are 0.2276, 0.1169, 0.1418, 0.0908, 0.0879, 0.0788, 0.1536, and 0.0809, respectively. These are feature-error scores from the visualization cell, not the final combined anomaly scores used for thresholding, so they should not be compared directly with $\tau = 0.100311$.

““latex

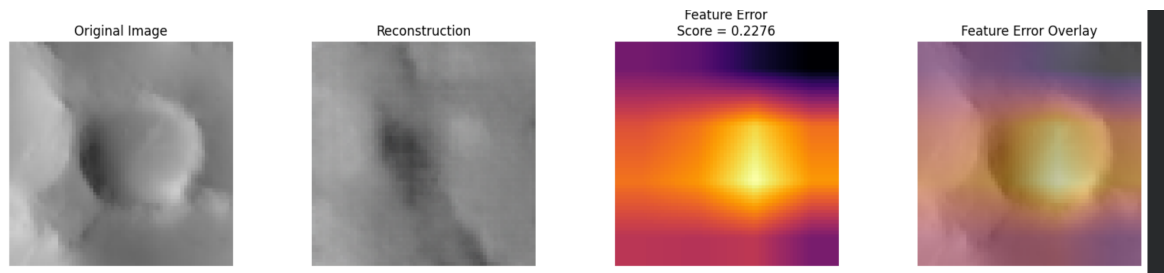


Figure 5: f-AnoGAN feature discrepancy visualization for normal example 1.

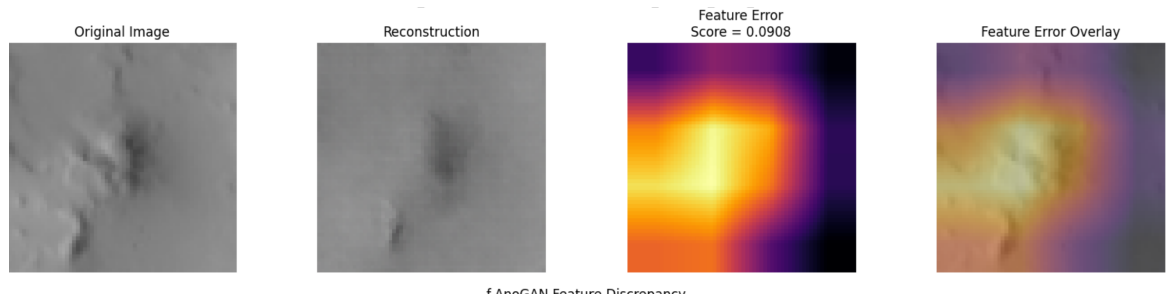


Figure 6: f-AnoGAN feature discrepancy visualization for normal example 2.

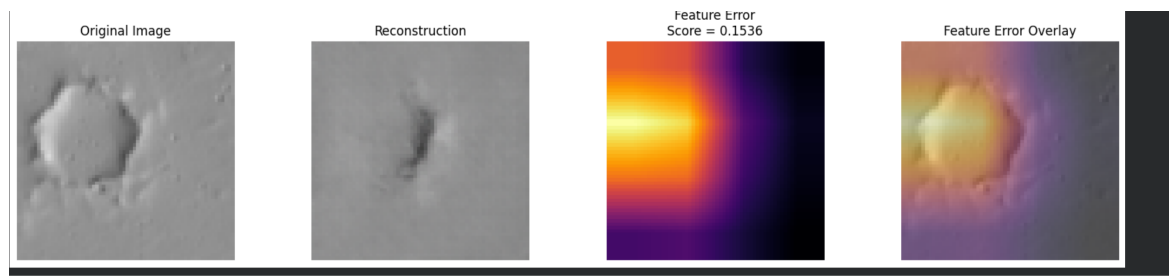


Figure 7: f-AnoGAN feature discrepancy visualization for normal example 3.

““

8 Model Comparison

The primary comparison uses AUROC because it does not depend on a particular operating threshold.

Version	Model	AUROC	Threshold
Version 1	CAE	0.9163	0.00823
Version 2	CVAE	0.8599	0.04929
Version 3	f-AnoGAN	0.9687	0.100311

Table 12: Current AUROC comparison after execution of the supplied f-AnoGAN notebook.

8.1 Interpretation

The currently available results change the interpretation of the model progression.

Version 1 achieves the highest AUROC among the two completed evaluations:

$$0.9163 > 0.8599.$$

Therefore, the CVAE cannot be described as an empirical improvement over the CAE on the current test results.

At the same time, the models use different preprocessing and scoring mechanisms. Version 1 operates at 224×224 and uses a maximum local reconstruction score, whereas the f-AnoGAN implementation operates at 64×64 and combines reconstruction and discriminator feature errors. Therefore, performance differences should be interpreted together with computational cost, resolution, and anomaly-score design.

The Version 1 result also shows why AUROC and thresholded metrics should be reported separately. Although AUROC is 0.9163, recall at the chosen threshold is only 0.3337. The threshold deliberately limits the normal flagged rate to approximately 5%, trading recall for precision.

9 Model Evolution: Symptom $\rightarrow$ Diagnosis $\rightarrow$ Fix

Transition	Symptom	Diagnosis	Fix / Evolution
v1 $\rightarrow$ v2	A deterministic autoencoder provides a useful reconstruction-based baseline, but a plain latent representation does not explicitly regularize the distribution of encoded samples.	A deterministic bottleneck does not impose a probabilistic prior on latent codes.	Introduce a variational bottleneck with μ , $\log \sigma^2$, stochastic sampling, and KL regularization.
v2 $\rightarrow$ v3	The CVAE is still reconstruction-driven and may trade sharpness against likelihood-style reconstruction objectives.	Adversarial training can provide a stronger learned realism signal, while discriminator features can provide an additional anomaly signal.	Adopt WGAN-GP for stable adversarial training, then train a dedicated encoder and combine pixel and discriminator-feature errors.

Table 13: Consolidated model-evolution log.

9.1 Why Version 3 Is Architecturally More Complex

The progression is not simply a sequence of larger networks:

1. CAE learns reconstruction.
2. CVAE adds probabilistic latent regularization.
3. f-AnoGAN adds adversarial manifold learning and a feature-space anomaly signal.

Thus the final model introduces two complementary notions of abnormality:

- **pixel disagreement:** the input cannot be reconstructed accurately;
- **feature disagreement:** the input and reconstruction are perceived differently by the learned critic.

10 Conclusion and Future Work

This work develops a progressive unsupervised anomaly-detection pipeline for HiRISE Martian surface imagery. The models evolve from a convolutional autoencoder to a variational autoencoder and finally to f-AnoGAN.

The newly evaluated Version 1 CAE achieves

$$\text{AUROC} = 0.9163$$

on the available real-anomaly test set. Its high AUROC indicates strong ranking performance, while its thresholded recall of 0.3337 highlights the practical importance of threshold selection.

The Version 2 CVAE achieves AUROC 0.8599 and therefore does not empirically outperform Version 1 on the currently available measurements.

Version 3 f-AnoGAN provides the most sophisticated architecture in the project: WGAN-GP learns a normal-image manifold, a dedicated encoder provides fast feed-forward inversion, and anomaly scores combine pixel and discriminator feature errors. Its executed final evaluation gives AUROC 0.9687 and AUPRC 0.9863, with F1 0.9400 at the calibrated threshold.

Future improvements include:

- evaluating f-AnoGAN at higher image resolutions such as 128×128 or 256×256 ;
- systematically tuning the anomaly-score weighting;
- comparing multiple threshold quantiles on a genuinely held-out normal calibration set;
- investigating category-wise recall for Sensor Artifact and Corrupted Crop;
- combining reconstruction, feature-space, and latent-space anomaly scores;
- measuring inference time and GPU memory alongside AUROC;
- addressing the non-Martian-content category explicitly if representative labeled evaluation data are available.

A Reproducibility

A.1 Version-Specific Configuration Summary

Version	Key configuration
CAE	224×224 RGB; batch 64; MSE; Adam; learning rate 10^{-3} ; weight decay 10^{-5} ; 10 epochs; local 28×28 average-pooling score followed by spatial/channel maximum; threshold = 95th percentile of normal test scores.
CVAE	Previously evaluated probabilistic convolutional autoencoder; 30 logged epochs; AUROC 0.8599; threshold 0.04929. Complete notebook-specific configuration is not reconstructed here without the Version 2 notebook.
f-AnoGAN	64×64 RGB; batch 32; latent dimension 128; WGAN-GP for up to 70 epochs with patience 15; Adam (2×10^{-4} , $\beta_1 = 0$, $\beta_2 = 0.9$); five critic updates per generator update; gradient penalty 10; encoder for 50 epochs with Adam (2×10^{-4} , 0.5, 0.999); score weight 0.9 reconstruction / 0.1 feature; calibration fraction 15%; threshold quantile 90%; calibration seed 42.

A.2 Recorded Version 1 Confusion Matrix

	Predicted Normal	Predicted Anomaly
Actual Normal	1993	105
Actual Anomaly	3415	1710

A.3 Important Reproducibility Notes

1. The Version 1 notebook trains at 224×224 , not 64×64 .
2. The Version 1 notebook uses batch size 64, not 32.
3. The Version 1 final anomaly score is a *patch-max* reconstruction score, not a global mean reconstruction error.
4. Version 1 threshold calibration uses the normal test scores at the 95th percentile; this differs from the normal-only training calibration used by the f-AnoGAN notebook.
5. f-AnoGAN uses 64×64 RGB input normalized to $[-1, 1]$, matching its Tanh generator output.
6. The executed f-AnoGAN notebook reports AUROC 0.9687, AUPRC 0.9863, accuracy 0.9157, precision 0.9498, recall 0.9303, F1 0.9400, and threshold 0.100311.

References

- [1] T. Schlegl, P. Seeböck, S. M. Waldstein, G. Langs, U. Schmidt-Erfurth, *f-AnoGAN: Fast Un-supervised Anomaly Detection with Generative Adversarial Networks*, Medical Image Analysis, 2019.